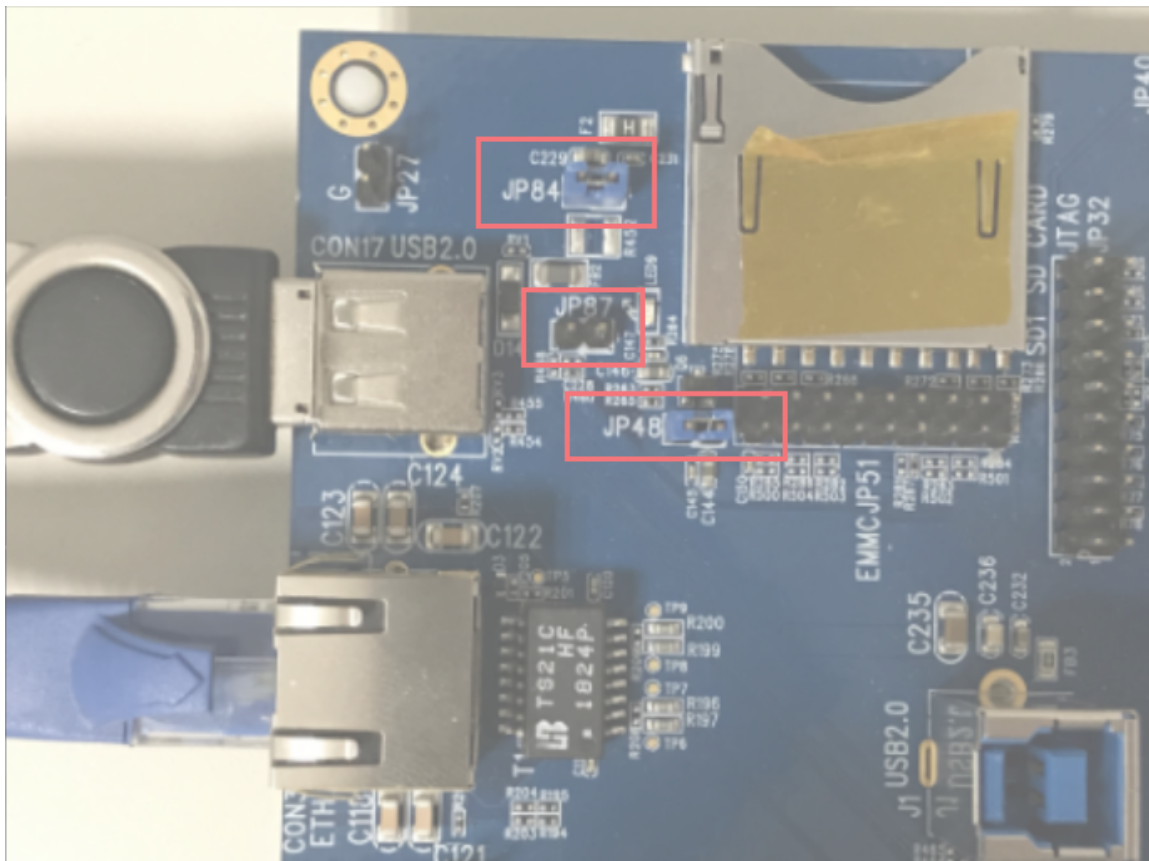


USB & SD UPGRADE

1. Hardware Connection

The EVB connection is as follows:



When using SD card to upgrade, connect JP48; when using USB to upgrade, the mainboard is host, connect JP84 and disconnect JP87.

2. SD Card Upgrade

SD card upgrade is enabled by default, and it supports blank upgrade. After the chip is powered on, if there are IPL, IPL_CUST and UBOOT in the SD card, they will be loaded and run.

2.1. Build SD Card Upgrade Package

After completing `make image` in the project, use

`make_sd_upgrade_sigmaster.sh` to package the SD card upgrade package.

```
tianhui.he@xmbc6401:~/codes/bringup_p3/project$ ./make_sd_upgrade_sigmaster.sh
Upgrade [[mxp? (y/N)y
Upgrade [[ipl? (y/N)y
Upgrade [[ipl_cust? (y/N)y
Upgrade [[uboot? (y/N)y
Upgrade [[kernel? (y/N)y
Upgrade [[rootfs? (y/N)y
Upgrade [[miservice? (y/N)y
Upgrade [[customer? (y/N)y
Upgrade [[appconfigs? (y/N)y
UpgradeImage Generating.....
1+0 records in
1+0 records out
16 bytes copied, 0.000113274 s, 141 kB/s
```

After packaging, `SigmastarUpgradeSD.bin` and `upgrade_script.txt` will be generated in `project/image/output/images`, as shown in the figure below. Copy these two files to the root directory of the SD card.

名称	修改日期	类型	大小
boot	2020/10/23 17:18	文件夹	
scripts	2020/10/23 17:50	文件夹	
appconfigs.jffs2	2020/10/23 17:18	JFFS2 文件	832 KB
auto_update.txt	2020/10/23 17:50	文本文档	1 KB
customer.jffs2	2020/10/23 17:18	JFFS2 文件	5,888 KB
dummy_pad	2020/10/23 17:50	文件	10 KB
ipl_cust_s.bin	2020/10/23 17:18	BIN 文件	30 KB
ipl_s.bin	2020/10/23 17:18	BIN 文件	25 KB
kernel	2020/10/23 17:18	文件	1,768 KB
miservice.sqfs	2020/10/23 17:18	SQFS 文件	4,936 KB
partition_layout.txt	2020/10/23 17:18	文本文档	1 KB
rootfs.sqfs	2020/10/23 17:18	SQFS 文件	1,724 KB
SigmastarUpgradeSD.bin	2020/10/23 17:50	BIN 文件	15,345 KB
uboot_s.bin	2020/10/23 17:18	BIN 文件	114 KB
upgrade_script.txt	2020/10/23 17:50	文本文档	16 KB

2.2. Flash Upgrade

To upgrade an empty flash, you need to copy `IPL.bin` , `IPL_CUST.bin` and `u-boot.xz.img.bin` in the boot directory to the SD card root directory, and rename them to `IPL` , `IPL_CUST` and `UBOOT` respectively. After the chip is powered on, they will be used to get a uboot environment that can be upgraded.

名称	修改日期	类型	大小
 IPL.bin	2020/10/23 17:18	BIN 文件	25 KB
 IPL_CUST.bin	2020/10/23 17:18	BIN 文件	30 KB
 MXP_SF.bin	2020/10/23 17:18	BIN 文件	4 KB
 u-boot.xz.img.bin	2020/10/23 17:18	BIN 文件	114 KB

2.3. Use SD Card To Upgrade In Uboot

Take out the SD card from the PC and insert it into the board, power on the board, press and hold Enter to enter the BootLoader command line, and execute `# sdstar` to upgrade the SD card.

Environment variables can be set by `setenv sdautoupgrade 1; saveenv` , and you can upgrade automatically if you have an SD card and an upgrade package.

3. USB Upgrade

The blank chip upgrade in USB is not supported, and the user needs to manually enable the USB upgrade function.

3.1. Enable USB upgrade function

Modify uboot config to enable USB upgrade function.

Use EVB defconfig to configure and then enable the following configuration by `make menuconfig` .

- Enable the following configs in Device Drivers

```
.config - U-Boot 2015.01 Configuration
Device Drivers

Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys.
Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
Legend: [*] built-in [ ] excluded <M> module <> module capable

[ ] Enable Driver Model
[ ] NAND Device Support --->
[ ] Enable watchdog
[*] Support for Host-side USB
*** USB Host Controller Drivers ***
[ ] xHCI HCD (USB 3.0) support (NEW)
[ ] EHCI HCD (USB 2.0) support (NEW)
[*] USB Mass Storage support
USB Gadget Support --->
DFU support ----
↓ (+)

<Select> < Exit > < Help > < Save > < Load >
```

- Enable the following configs in MStar drivers

```
.config - U-Boot 2015.01 Configuration
Device Drivers Mstar drivers

Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys.
Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
Legend: [*] built-in [ ] excluded <M> module <> module capable

--- Mstar drivers
[*] MSTAR ISP NOR FLASH
[*] Save environment to ISP NOR FLASH
[ ] MTD device for ISP NOR FLASH
[*] Enable Mstar partition support
[*] MSTAR SDMMC
[ ] MSTAR eMMC
[*] MSTAR Gpio
[ ] MSTAR NAND
[ ] EYE
↓ (+)

<Select> < Exit > < Help > < Save > < Load >
```

```
.config - U-Boot 2015.01 Configuration
Device Drivers Mstar drivers

Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys.
Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
Legend: [*] built-in [ ] excluded <M> module <> module capable

--- Mstar drivers
[*] MSTAR USB
[*] MSTAR EMAC
[ ] MSTAR EMAC 1
[*] Supply to internal PHY
[ ] Supply to RMII
[ ] Supply to IC+ Phy
[ ] EMAC fix link to mii/rmii
[*] Enable 1st EHC
[*] Enable 2nd EHC
[*] Enable 3rd EHC (NEW)
↓ (+)

<Select> < Exit > < Help > < Save > < Load >
```

3.2. Pack The USB Upgrade Package

Similar to [2.1 Build SD Card Upgrade Package](#) [#21-Build-SD-Card-Upgrade-Package], after running `make_usb_upgrade_sigmaster.sh` in the project, `SigmasterUpgrade.bin` and `upgrade_script.txt` will be generated in `project/image/output/images`, copy them to the USB root directory.

3.3. Upgrade With USB In Uboot

Boot into uboot, set the usb upgrade port to 0:

```
filesize=110
ipaddr=172.19.24.77
netmask=255.255.255.0
serverip=172.19.24.105
sf_kernel_size=220000
sf_kernel_start=50000
sf_part_size=d0000
sf_part_start=f30000
stderr=serial
stdin=serial
stdout=serial
usb_folder=images

Environment size: 727/4092 bytes
SigmaStar # setenv UpgradePort 0
SigmaStar # save
Saving Environment to SPI Flash...
env offset=0x4F000 env size=0x1000
Flash is detected (0x0709, 0xEF, 0x40, 0x18)
SF: Detected nor0 with total size 16 MiB
Erasing SPI flash..._spi_flash_erase: addr 0x4f000, len 0x1000 (cost 43 ms)
Writing to SPI flash..._spi_flash_write to 0x4f000, len 0x1000 from 0x23b0f9b0 100%(cost 25 ms)
done
SigmaStar # usbstar
```

Execute usbstar for USB upgrade, and it will automatically restart after that.